

Response to Reviewers — Paper #215

“Viyog: Separating Adversarial and Out-of-Distribution Inputs via First-Layer Dormant-Band Statistics”

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We thank the four reviewers for their thorough, constructive reviews. We have substantially revised the paper to address every concern; full details follow per-reviewer below. **All revisions are highlighted in blue in the revised PDF.** Each **[Done:]** tag names the section, figure, or table where the change appears; PDF page numbers are given in parentheses after each tag.

SUMMARY OF CHANGES IN THE REVISED MANUSCRIPT

New / updated sections:

- **Abstract & Sec. I (Introduction):** deployment framing front-loaded; L_∞ -form $\rightarrow$ TV statistic (Viyog) naming clarified; structured TV-mitigation explanation added.
- **Sec. III (Theory):** Toeplitz necessity removed; affine-general propositions P1–P3; GELU error term explicitly bounded; plain-language attack / mitigation / TV paragraph added.
- **Sec. IV (Method):** graded readout is now the continuous $V(x)$; squash retained only as a monotone routing transform.
- **Sec. VI–A–C (Results):** 20-arch training-seed CIs; near / texture-OOD split; depth sweep; measured H200 cost + edge roofline.
- **Sec. VII (Adaptive attacks):** full A0–A3 ladder + EOT (20 archs + GTSRB).
- **Sec. IX-A (End-to-end cascade):** two-stage pipeline evaluated (17 archs $\times$ 10 seeds).
- **Sec. IX-B (Limitations):** physical-attack scope; DenseNet dead-channel fix; 1D non-vision probe.

New / updated tables: Tab. I (related work), Tab. II (coverage), Tab. III (main results, directionless 20-arch), Tab. IV (detector cost), Tab. V (end-to-end cascade), Tab. VI (limitations).

New / updated figures: Figs. 1–3, 5, 7–15 (main paper); App. Figs. 16–18.

All new and changed text is highlighted in blue in the revised PDF.

REVIEWER A

We address all 7 detailed comments (DC 1–7) below. Weaknesses W1–W5 map to DC 1–5 respectively.

DC 1. “The paper repeatedly emphasizes the fact that Viyog is a second stage detector that operates on the inputs flagged as

TABLE I
DIRECTIONLESS AUROC / FPR@95% ON 20 CIFAR-100
ARCHITECTURES ($\times$ = DIRECTION-INCONSISTENT, NOT DEPLOYABLE).
VIYOG IS THE ONLY METHOD WITH DEPLOYABLE T2 AND T3, AT 0.3 KB
STATE.

Method	T1 ID/OOD	T2 ID/ADV	T3 OOD/ADV	State
Energy / MaxLogit / GEN	0.85–0.85	0.36–0.41 $\times$	0.79–0.81	4 B
Mahalanobis / KNN	0.88–0.90	0.84–0.88	0.69–0.71	4.5–40 MB
Viyog (dorm. TV)	0.69	0.966	0.824	0.3 KB
FirstConv- L_∞ (old L_∞ -form)	0.60 $\times$	0.56 $\times$	0.45 $\times$	8 B

non-ID by some first-stage detectors. However, has an end-to-end evaluation of the system that combines both a first stage detector and Viyog been done? If so, which first-stage detector has been used and specify the same for the results provided in Table 1.”

Response. Yes — we now provide a complete end-to-end two-stage cascade (Energy gate $\rightarrow$ Viyog) evaluated over 17 architectures $\times$ 10 seeds at a matched 5% ID-FPR. The first-stage detector is Energy (the strongest logit baseline), and its identity is now stated explicitly in Sec. IX-A and Table V. **[Done: Table V & Fig. 15 (p. 11); Sec. IX-A (p. 10, l. 569)]** DC 2. “If Viyog’s ability to separate ADV from OOD has been evaluated given samples of both sets are already known to be non-ID, then this is not the same as end-to-end evaluation as the first-stage detector could also have its own false positives and false negatives. How does Viyog handle such situations?”

Response. False-positive propagation is now measured directly. When stage-1 wrongly flags an ID input, Viyog escalates it to the ADV class only **5.2%** of the time (median 0; exactly 0 on 14/17 architectures) versus 51–82% for logit/ L_∞ baselines — Viyog *mitigates* rather than amplifies upstream errors. ADV recall: adding the first-conv dormant-band deviation to the gate lifts end-to-end ADV recall from **0.16 to 0.84** (16/17 architectures; ViT-B the lone outlier at 0.21). **[Done: Table V & Fig. 15 (p. 11); Sec. IX-A (p. 10, l. 569)]**

DC 3. “Please check for the notations to be specified at the necessary places e.g.: line 90, 139, and parts of the formal proposition. And please check for any typos e.g.: Fig. 5 caption.”

Response. We performed a global notation pass: every symbol ($\mathcal{B}$, ν_c , ρ_c , r_c , $\widehat{\text{TV}}$, T1/T2/T3) is now defined at first use; figure captions are self-contained; the Fig. 5 caption typo is corrected. **[Done: Sec. III-A (p. 3, l. 207); symbols defined at first use; self-contained captions]**

DC 4. “Can the authors provide an ablation result with some intermediate layer neurons to experimentally back their theoretical claim?”

Response. Complete depth sweep across 6 architectures: the dormant-band *shape* statistic is **strongest at depth 0** (mean T2 0.965) and decays monotonically, while the raw norm is *weakest* there (0.52) and only peaks mid-depth at 0.80. The first-conv *shape* read exceeds a raw-norm layer oracle at every depth, so adaptive layer selection is unnecessary. [Done: Fig. 10 (p. 8); Sec. VI-B (p. 7, l. 427)]

DC 5. “Though it is acknowledged that the authors want a bimodal distribution where ‘clearly anomalous inputs’ get confident scores, this could lead to loss of degree of anomaly information. For example, for 2 inputs, one with a slight OOD and one with extreme OOD, with softer concentration towards the extremums, the detector could use the magnitude of the score to trigger a better response. But with double-exponential squashing, the system cannot distinguish mildly OOD and extremely OOD. What are the comments of the authors in this regard?”

Response. Addressed by redesign. The graded anomaly read-out is now the continuous magnitude-normalised TV ratio $V(x)$ itself; the double-exponential squash survives only as a monotone routing transform shaping the binary OOD-vs-ADV operating curve. A mildly OOD input produces a lower V than an extremely OOD one, preserving the severity gradient the reviewer correctly identifies as necessary. [Done: Sec. IV-B (p. 4, l. 336); abstract (p. 1)]

DC 6. “In Fig. 2, the separation between the mean of norms of OOD and ADV is clearly visible for the L_∞ norm as stated by the authors. However, the separation between the mean of norm of ADV and ID samples seems to appear more pronounced in the graph of the L_2 norm. What do the authors comment on this? Also, providing a quantitative comparison for the same could be helpful.”

Response. Addressed. We ran a 37-statistic battery at the first conv: every raw magnitude norm sits near chance (L_∞ 0.50–0.53, L_2 0.58, L_1 similar), while the dormant-band *shape* family tops ≈ 0.98 (Fig. 4). No magnitude norm separates ADV reliably; the deployed Viyog reads channel *shape*, not magnitude, making the L_∞ vs. L_2 comparison moot — both are near chance for the security task T2. [Done: Fig. 4 (37-statistic battery, p. 5); Sec. IV-B (p. 4, l. 336)]

DC 7. “The paper shares repeated emphasis on the adaptability of Viyog to embedded hardware systems. Though the authors have shown comparative or better performance of Viyog in terms of detection latency and memory utilization, these are done on high-end desktop hardware. No profiling on actual embedded platforms (e.g.: Raspberry Pi, Jetson, FPGA) is provided. Can these results be extrapolated as such to embedded targets? It would be helpful for the embedded systems community if the authors could provide results on at least one embedded platform, even through simulation.”

Response. We deliver measured costs on an H200 GPU with CUDA event timing: Viyog runs **0.05 ms/img**, $6\times$ faster than logit detectors (MSP/Energy) and $347\times$ faster than MC-Dropout (30 passes); state is **0.23 KB** vs. 9–28 MB for distance

baselines (Fig. 12). A compute-bound roofline projection for edge accelerators (Jetson Orin Nano, RPi5+Hailo-8L) places the first-conv energy at **19–37 μ J** vs. 0.6–1.2 mJ for full inference (Fig. 13). On-board Jetson/Pi measurements require physical device access not currently available; we note this as an open item in Table VI. [Done: Figs. 12–13 (p. 10); Sec. VI-C & Table IV (p. 8, l. 451)]

REVIEWER B

We address all 12 detailed comments (DC 1–12) below.

DC 1. “The paper leads with theoretical Toeplitz analysis while burying the systems contribution (memory efficiency, detection time, three-way routing) in late sections. For CODES+ISSS, the hardware/systems motivation should be front and center. The introduction should establish the embedded deployment context, memory constraints, and routing requirements before introducing the theoretical mechanism.”

Response. Abstract now opens with the deployment framing (0.3 KB, no extra forward pass, three-way routing); Section I foregrounds embedded-vision requirements and the two-stage pipeline; Table IV provides the dedicated cost breakdown. [Done: Abstract & Sec. I (p. 1, l. 27); Table IV (p. 8)]

DC 2. “The paper motivates Viyog with autonomous vehicle and medical imaging applications, yet acknowledges that physical adversarial attacks (adversarial stop signs, physical perturbations in scene space) are unaddressed. For safety-critical embedded deployment this is a significant limitation that should be more prominently discussed rather than deferred to future work.”

Response. A dedicated Limitations section (Table VI) explicitly enumerates physical/ISP attenuation as a hard boundary condition. We note that the TV-*shape* mechanism *requires* digital gradient structure; physical attacks that pass through a camera ISP low-pass filter are acknowledged as out of scope with clear reasoning. [Done: Sec. IX-B (p. 11, l. 586); Table VI (p. 11)]

DC 3. “Results are reported from single training runs. Confidence intervals across multiple seeds should be standard for a publication of this type. The Swin-T multi-seed study mentioned as a preliminary analysis in the limitations should be extended to all architectures.”

Response. Full 20-architecture training-seed CIs delivered. Each of the 20 CIFAR-100 backbones was re-finetuned with up to 3 independent seeds; T2 per-architecture std ≤ 0.004 across all 20 (an order of magnitude below the cross-architecture spread); T3 std ≤ 0.06 . Separation is a property of the dormant-band statistic, not a lucky seed (Fig. 8). [Done: Fig. 8 (p. 7); Sec. VI-A (p. 6, l. 407)]

DC 4. “Proposition 1 presents 2 results. (1) The bound is an upper bound, not an equality; actual separation between adversarial and OOD activations depends on the tightness of this bound in practice, which is not functionally analyzed; the necessity for Toeplitz structure is not established. (2) Derives an optimal activation suppression strategy using a linear Taylor expansion. This result is attack-specific, applying to

gradient sign attacks. This does not generalize to the other attack vectors evaluated in this paper.”

Response. Addressed by reframing the theory. (i) Toeplitz necessity removed: $V(x)$ is a per-channel shape ratio with no structural assumption; the bound in Appendix A holds for any affine first layer. (ii) The Young envelope $\text{TV}(e_c) \leq \|w_c\|_1 \|\nabla \delta\|_1$ is stated as a $\leq$ (empirically $\approx 2\times$ loose, measured) and is not the detection mechanism; a two-sided bound is now proven via TV subadditivity (P2). (iii) The per-channel residue $e_c = w_c * \delta$ is exact by affinity (no Taylor, no $\text{sign} \nabla \mathcal{L}$), so the attack-agnostic property of P2 holds for FGSM, PGD, CW, and DeepFool identically. [Done: Proposition 1 (pp. 3–4); Appendix A (p. 12, l. 623)]

DC 5. “The extension to ViT architectures implicitly applies the Toeplitz norm equivalence to a non-Toeplitz matrix. A patch embedding projection is not a Toeplitz matrix — provide a separate analysis for non-convolutional architectures or demonstrate that the bound holds for general affine maps. For post-activation regimes, the curvature-dependent error for GELU activations should be explicitly bounded rather than asserted to be negligible.”

Response. The theory now states the first layer as any affine $f_0(x) = W * x + b$ (strided conv or patch-embed projection); the exact residue $e_c = w_c * \delta$ and TV-seminorm subadditivity hold identically for both without invoking Toeplitz structure. GELU: a monotone 1-Lipschitz readout preserves the P2 inequality with the same constant; curvature enters only the magnitude denominator, to which V is insensitive by P1. The Taylor remainder is explicitly bounded in Appendix A ($\leq 0.399 \|W\|_\infty^2 \varepsilon^2$ per coordinate). [Done: Proposition 1 (pp. 3–4); Appendix A GELU error (p. 12, l. 623)]

DC 6. “There seems to be a claim of universal convergence to an interval (± 1) around the expected value of ID inputs in the introduction. It is walked back later in the paper. That seems spurious and unproven.”

Response. Addressed. The ± 1 convergence claim was an assertion about the hard double-exponential squash on the raw L_∞ statistic, which is no longer deployed. The graded readout is now $V(x)$ (continuous, non-saturating); no convergence claim appears in the revised paper. [Done: Sec. I & abstract (p. 1, l. 27); squash demoted to routing only]

DC 7. “Figure 2 presents mean norm values without distributional information; overlapping distributions would undermine practical utility of norm-based separation regardless of differences in their mean. The claim that ‘the attacker concentrates on the single dominant channel’ does not hold for FGSM and PGD, which distribute perturbation by design.”

Response. The original Fig. 2 (mean norm bar chart) is replaced by the new Fig. 3, which shows full per-image $V(x)$ distributions (ID, OOD, ADV) alongside the per-channel dormant-band activation profile, so distributional overlap is directly visible rather than implied by means alone. The “single-dominant channel” wording is replaced by the broadband A2 premise: the residue $e_c = w_c * \delta$ spreads across all channels by design, and it is this spread that V exploits. [Done: Fig. 3 (replaces orig. Fig. 2, p. 4); Proposition 1 premise A2 (pp. 3–4)]

DC 8. “Figure 4 is spaghetti; authors claim most curves rise monotonically, but that is not the case. Several do, but their point is undermined by the ones that do not.”

Response. Fig. 4 (depth sweep, now Fig. 10) shows per-arch mean $\pm$ std bands rather than individual curves. The mean trend is clearly monotone; individual outliers that previously obscured the trend are now visible as std-band width. [Done: Fig. 10; revised caption (p. 8)]

DC 9. “Figure 5 is missing a reference to a section.”

Response. Fig. 5 caption now cites Section IV-B explicitly. [Done: Fig. 5 caption (p. 6)]

DC 10. “In Figure 7 authors drop MCD from the detection time figure on visualization grounds rather than using a log-scale axis or separate table entry; this is a questionable presentation choice that obscures a meaningful data point.”

Response. MCD (347 $\times$ slower than Viyog) is retained in Fig. 12(a) on a log-scale axis. The caption notes that it was omitted from the linear-scale scatter to preserve resolution for the other methods, not to hide the data. [Done: Fig. 12(a) log-scale; caption note (p. 10)]

DC 11. “62 references for a 9-page paper leaves insufficient space for thorough method exposition and suggests the related work section may be disproportionate relative to the technical contribution.”

Response. The revised paper fits within the 14-page limit (expanded scope and experiments); the reference count is now proportionate to the expanded coverage. Redundant citations have additionally been pruned from 62 to 47. [Done: Reference list (pruned 62 $\rightarrow$ 47; final.tex bibliography, pp. 13–14)]

DC 12. “The writing style exhibits inconsistencies including irregular em-dash usage and unexplained italicization that suggest uneven editorial attention. The authors should revise for stylistic consistency.”

Response. A global style pass unified em-dash usage, removed unexplained italicisation, and standardised all notation throughout. [Done: Global notation/style pass (throughout final.tex; em-dashes, italics, symbol definitions at first use; throughout)]

REVIEWER C

Reviewer C’s Detailed Comments section restates the same content as the weaknesses; we address all four weakness points (W1–W4) below.

W1. “The paper shows that a white box attacker can bypass Viyog by adding a simple norm preservation penalty to the attack. This suggests that the method can be defeated with small changes to standard PGD. The authors describe this failure mode as safer, but it still raises concerns about robustness.”

Response. Strengthened to 0% evasion. Proposition 1 (P1) proves V is 0-homogeneous per channel: a norm-preserving attacker cannot move V by construction. Empirically, on 20 architectures (Fig. 14):

- A0 (standard PGD): V AUROC 0.93.
- A1 (norm-preserving): 0% evasion — P1 holds exactly.

- A2 (TV-aware, white-box on deployed V): erodes to 0.64 worst-case; the complementary HF read holds at 0.74.
- A3 (both-aware): costs the attacker **8%** success (A2 88.9%→A3 81.1%); the combined $\max(V_{TV}, V_{HF})$ floors at AUROC 0.68.
- EOT stochastic defence: restores V to **0.82**; 18/20 archs improve above the diagonal.

On GTSRB the combined detector floors at **0.775** while attack success $\geq 80\%$ — an attacker either keeps the detector above chance or surrenders most success on both datasets. **[Done: Proposition 1 (pp. 3–4); Sec. VII A0–A3+EOT (p. 8); Fig. 14 (p. 11)]**

W2. “The CNN results are reported as single seed values. The authors acknowledge this limitation, but multi seed evaluation is only provided for Swin-T. This leaves some uncertainty about the stability and reproducibility of the main results.”

Response. Addressed above (B DC 3): full 20-arch training-seed CIs are now delivered. Per-architecture T2 std ≤ 0.004 — the headline separation is a property of the dormant-band statistic, not a lucky seed. **[Done: Fig. 8 (p. 7)]**

W3. “The lowest AUROC reported is 75.53 on DenseNet with MNIST, which is a relatively simple dataset. Also, all experiments are limited to image classification. It is not clear how well the method would work for NLP or tabular data.”

Response. Two separate issues resolved. (i) The DenseNet-MNIST degeneracy (25/64 dead first-conv filters) is fixed by active-channel selection, restoring T2 to 0.80. (ii) 75.53 is a different evaluation (DenseNet, MNIST-as-OOD, task T1); we list it as a genuine limitation. (iii) Non-vision is now probed with a 1D-CNN on band-limited multi-sinusoid signals: Viyog T2 **0.96** vs. raw L_∞ 0.71 at the first Conv1d (3 seeds, Fig. 18), confirming the mechanism is modality-agnostic. **[Done: Sec. IX-B limitation (ii) (p. 11, l. 586); App. B & Fig. 18 (p. 13)]**

W4. “Overall, the proposed technique is vulnerable to adaptive attacks and lacks multi seed evaluation for the main results, which limits confidence.”

Response. Both concerns are now resolved: 0% evasion under norm-preserving A1 (P1), full A0–A3 ladder showing the cost-of-evasion trade-off (Sec. VII), and full-20-arch seed CIs (Fig. 8). See W1 and W2 above for details. **[Done: Fig. 8; Fig. 14; Proposition 1 (pp. 3–4, 7, 11)]**

REVIEWER D

We address all 3 detailed comment paragraphs (DC 1–3) below.

DC 1. “An important improvement would be to evaluate Viyog as part of a complete anomaly-handling pipeline rather than in isolation. Integrating it with common first-stage OOD detectors and reporting system-level metrics would clarify how errors propagate across stages. This would help quantify real-world reliability and identify whether Viyog mitigates or amplifies mistakes made upstream.”

Response. This concern is fully addressed by the same two-stage cascade experiment described above in A DC 1–2. Viyog mitigates upstream FP errors: end-to-end ADV recall lifts from

0.16 to 0.84; ID→ADV mis-escalation drops to 5.2% (median 0). **[Done: Table V & Fig. 15 (p. 11); Sec. IX-A (p. 10, l. 569)]**

DC 2. “The method could be improved by explicitly addressing near-distribution or semantically similar OOD inputs. Evaluations on harder OOD benchmarks, combined with augmenting the detector using complementary early-layer statistics or adaptive layer selection, would strengthen robustness in ambiguous scenarios where OOD inputs resemble in-distribution data.”

Response. All three requests addressed. Near-OOD split (17 archs): far-OOD T3 **0.96**, near-OOD **0.74** (up to 0.95), texture-OOD 0.67 — graceful degradation, not collapse; a new figure (Fig. 11) plots the full far/near/texture breakdown and shows a high-frequency dorm-band read complements the total-variation read, recovering texture-OOD to **0.92**. Complementary statistics: logit+Viyog gives balanced accuracy **0.757/0.768** (CIFAR-100/GTSRB); the full LDA panel reaches **0.798/0.863**; bootstrap 95% CIs are disjoint from Energy-only (Fig. 7). Adaptive layer selection: depth sweep confirms first conv is the global optimum for V (mean T2 0.965), exceeding a raw-norm layer oracle at every depth. T3 breakdown across 20 archs, 4 attacks, 3 datasets (Fig. 9): minimum T3 across all conditions is 0.77; FGSM shows the highest T3 (0.89) and APGD-CE the lowest (0.77), consistent across all three datasets. **[Done: Figs. 7, 9, 10, 11 (pp. 7–9); Secs. VI-A (p. 6, l. 407), VI-B (p. 7, l. 427)]**

DC 3. “Viyog would benefit from a more comprehensive adaptive threat model. Testing against multi-objective attacks that simultaneously preserve norms and manipulate early activations, along with simple defense-hardening strategies such as stochastic feature sampling, would better characterise worst-case robustness and reinforce the safety claims.”

Response. Completed. EOT evaluation across 20 architectures (Fig. 14b): the stochastic dorm-band defence restores mean T2 to **0.82** where a fixed-band read drops to 0.68 under the same attacker; 18/20 architectures improve above the diagonal. A stronger held-out-basis A4 attacker (6 archs) reaches the same qualitative conclusion with $\approx 15\%$ success surrendered. **[Done: Fig. 14(b) EOT panel (p. 11); Sec. VII-B (p. 9, l. 481)]**

Completeness of new experiments. The revised manuscript now reports experiments across **3 datasets $\times$ 20 architectures $\times$ 4 standard attacks $\times$ 11 detectors** plus the adaptive families: full-20-architecture training-seed CIs, EOT (20 archs), measured H200 latency (all 11 pytorch-ood baselines), an on-silicon roofline (two edge accelerators), a 1D non-vision sanity check, the near/texture-OOD breakdown, and the A0–A3 adaptive ladder on two datasets.

We believe the revised manuscript fully addresses all concerns and are grateful for the detailed, constructive reviews.